Gestalten

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0.1 Dualizability Composes

Let $A\colon \mathsf{StRing}$ and $B_n \in \mathbf{Alg}_{\mathcal{E}^\infty}(A_{n+1})$. Then we get an adjunction

$$\begin{tikzcd} A_{n+1} \arrow[r, bend left, "f^* = - \otimes B_n"] \arrow[r, phantom, "\perp"] & \mathbf{Mod}_{B_n}(A_{n+1}) \arrow[l, bend left, "f_*"] \end{tikzcd}$$

When does this adjunction preserves/reflects notions of dualizability? How about in more/less affine cases?

0.2 Rigidity

We fix $A\colon \mathsf{StRing}$. Note that for A 1-affine and $B_1 \in \mathbf{Alg}_{\mathcal{E}^\infty}(A_2)$, rigidity of the B_1 is equivalent to $f\colon A \to B$ and $\Delta_f\colon A \to B$ being 1-prim (Cor. 4.45 in [Ram24]).

Definition 0.1 ((Local) Rigidity)**.** Let $B_n \in \mathbf{Alg}_{\mathcal{E}^\infty}(A_{n+1})$ and the induced $f\colon A \to B$. We say that B_n is *rigid* if f is n-proper (I think the higher m conditions are redundant here). We could define a weaker notion of *k-local rigidity* by only asking that $\Delta_f^{(k)}, \Delta_f^{(k+1)}, \ldots$ are n-prim and that B_n is dualizable as a $\int_{S^{-1}} B_n$-module, $\int_{S^0} B_n$-module, $\ldots$, $\int_{S^{k-1}} B_n$-module.

Then hopefully, local-rigidity implies rigidity (which is -1-local rigidity) BUT one needs to find a non-affine definition to make this interesting.

1 Dualizability

Definition 1.1 (n-prim)**.** $f\colon A \to B$ is n-prim if, for all $m \geq n+1$, we have

$$\begin{tikzcd} \mathbb{1}_m \arrow[r, bend left, "f^\star_{m-1}"] \arrow[r, phantom, "\perp"] & (B/A)_m \arrow[l, dashed, bend left, "f_{m-1,\star}"] \end{tikzcd}$$

Definition 1.2 (n-proper)**.** $f\colon A \to B$ is n-proper if f, Δ_f, Δ_{Δ_f}, ... are all n-prim.

Definition 1.3 (n-suave)**.** $f\colon A \to B$ is n-suave if, for all $m \geq n+1$, we have

$$\begin{tikzcd} (B/A)^!_m \arrow[r, bend left, "f_{m-1,!}"] \arrow[r, phantom, "\perp"] & \mathbb{1}_m \arrow[l, dashed, bend left, "f^!_{m-1}"] \arrow[r, bend right, "f^\star_{m-1}"'] \arrow[r, phantom, "\perp"] & (B/A)_m \arrow[l, dashed, bend right, "f_{m-1,\#}"'] \end{tikzcd}$$

in A_{m+1}, where $(B/A)^!_m \to \mathbb{1}_m$ is the counit of the $f_{m,\#} \dashv f^\star_m$ adjunction in A_{m+2}.

Definition 1.4 (n-étale)**.** $f\colon A \to B$ is n-étale if f, Δ_f, Δ_{Δ_f}, ... are all n-suave.

Remark 1.1 (Redundancy)*.* For $m \geq n+1$, we have $(B/A)^!_m \simeq (B/A)_m$ and $f^\star_m \dashv f_{m,\#}$. In this range, we can denote $f_{m,\star} = f_{m,\#}$ as it lifts the external right adjoint (which always exists but isn't necessarily A-linear).

1.1 n-proper is useful to understand diagonals and tensoring

Remark 1.2 (Higher diagonals)*.* We get functors $(B \otimes_A B)_{m+1} \to \mathbf{End}_{A_{m+2}}((B/A)_{m+1})$.

Lemma 1.1

If $f\colon A \to B$ is n-prim, then $(B/A)_m \otimes_{A_{m+1}} (B/A)_m \simeq (B \otimes_A B/A)_m$ for $m \geq n$.

Proof: We can internally describe $(B/A)_m = f_{m,\star} \circ f^\star_m$. A Eckmann-Hilton argument then shows that $(B/A)_m \otimes_{A_{m+1}} (B/A)_m \simeq \mathbf{End}((B/A)_{m+1} \otimes_{A_m} (B/A)_{m+1})$. Hence "spectrification" only changes $(B/A)_m \otimes_{A_{m+1}} (B/A)_m$ for $m < n$. ■

Corollary 1.1.1

If $f\colon A \to B$ and Δ_f is n-prim, then $(B/A)_m$ is its own dual for $m \geq n+1$.

Proof: Similar to showin that rigid implies frobenius implies self-dualizable. The Beck-Chevalley condition comes from Δ_f being n-prim (the adjoint is $B \otimes_A B$-linear). ■

Conjecture 1.1. If $f\colon A \to B$ is n-proper, then the map $\int_{S^k} (B/A)_m \simeq S^k \otimes (B/A)_m \to (B/A)_m$ induced by the terminal $S^k \to \{\bullet\}$ is $(\Delta^{(k+1)}/A)_m$.

Pablo: the result above is nice, but etaleness will give an easier way to describe duality I think

Proof: If f is n-proper, then $A \to B \otimes_A B$ is n-prim. I **think** this should imply that tensoring internally in $(B \otimes_A B)_{m+1}$ is the same as tensoring internally in A_{m+1} over $(B \otimes_A B/A)_{m+1}$. Then this should imply the result by induction on k. ■

References

[Ram24] Maxime Ramzi. Locally Rigid ∞-Categories. 2024.